

Knowledge Distillation for Edge-Efficient Plant Disease Classification

Abdulkaki Celtik¹ and Meric Kelemeden¹

Department of Computer Engineering

Abstract. Accurate image-based plant disease classification can support precision agriculture, but high-capacity convolutional networks are not always suitable for edge or mobile deployment. This work evaluates whether knowledge distillation can transfer the predictive behavior of a ResNet50 teacher to a compact MobileNetV3-Small student on the 38-class PlantVillage dataset. We train a supervised ResNet50 teacher, a supervised MobileNetV3-Small baseline, and a distilled MobileNetV3-Small student using softened teacher predictions. On the held-out test split, the distilled student reaches 0.9937 accuracy and 0.9907 macro-F1 with 1.56M parameters, compared with 23.59M parameters for the ResNet50 teacher. These results indicate that, in this controlled dataset setting, distillation can preserve near-teacher classification performance in a model that is about 15.1 times smaller.

Keywords: Knowledge distillation · Plant disease classification · MobileNetV3 · ResNet50 · Edge inference.

1 Introduction

Plant disease recognition is a practical computer vision problem with direct relevance to crop monitoring and precision agriculture. Deep convolutional neural networks can achieve strong classification accuracy on curated leaf image datasets, but models with high parameter counts may be inconvenient for mobile or edge-oriented deployment. This creates a tension between predictive accuracy and computational efficiency.

This project studies that trade-off through knowledge distillation. A high-capacity ResNet50 teacher is trained on the full PlantVillage dataset and then used to guide a lightweight MobileNetV3-Small student. The aim is not to claim deployment on a specific device, but to evaluate whether a compact student can retain the teacher’s behavior while reducing parameter count and maintaining low batch-one inference latency.

The contributions are as follows. First, we train a ResNet50 teacher on the full 38-class PlantVillage task. Second, we train a supervised MobileNetV3-Small baseline under the same split and preprocessing pipeline. Third, we distill the ResNet50 teacher into MobileNetV3-Small using hard labels and softened teacher logits. Finally, we compare accuracy, macro-F1, parameter count, and measured latency across the teacher, supervised student, and distilled student.

2 Background and Related Work

PlantVillage provides a widely used collection of labeled plant leaf images for plant health research [4]. The dataset has enabled controlled studies of image-based plant disease classification, although its curated imaging conditions are cleaner than many field scenarios.

Knowledge distillation was introduced as a way to transfer information from a larger teacher model to a smaller student model [2]. Instead of learning only from one-hot labels, the student also learns from the teacher’s softened class probabilities, which can encode similarities between classes. This makes distillation attractive when the deployment model must be smaller than the model used to generate the strongest predictions.

The teacher in this work is ResNet50, a residual network architecture that uses skip connections to ease optimization of deeper convolutional models [1]. The student is MobileNetV3-Small, an architecture designed for mobile inference through efficient convolutional blocks and hardware-aware design choices [3]. The combination is therefore natural for an edge-oriented compression experiment: ResNet50 provides a strong teacher, while MobileNetV3-Small provides a compact student model.

3 Methodology

Let z_t and z_s denote the teacher and student logits for an input image, and let y denote the ground-truth class label. The supervised teacher and supervised student baselines are trained with cross-entropy against the hard labels. For distillation, the teacher is first trained and then frozen. The student is optimized using a weighted sum of the cross-entropy loss and a temperature-scaled Kullback-Leibler divergence term:

$$\mathcal{L} = \alpha T^2 \text{KL} \left(\text{softmax} \left(\frac{z_t}{T} \right) \parallel \text{softmax} \left(\frac{z_s}{T} \right) \right) + (1 - \alpha) \text{CE}(y, z_s). \quad (1)$$

In the final experiment, the distillation temperature is $T = 4.0$ and the distillation weight is $\alpha = 0.5$. The T^2 factor follows the common practice of compensating for gradient scaling introduced by softened probabilities. This notation matches the implementation used in the active distillation training script, where α controls the weight of the teacher-guided term.

MobileNetV3-Small is used as the student because it has substantially fewer parameters than ResNet50 while retaining a convolutional design appropriate for image classification. This makes it a suitable candidate for studying whether teacher supervision can preserve classification performance under a smaller parameter budget.

4 Experimental Setup

Experiments use the full PlantVillage directory available in the repository. The pipeline creates deterministic train, validation, and test splits with seed 42.

Images are resized to 224×224 pixels and normalized with ImageNet statistics. Training uses random horizontal flipping and small random rotations; validation and test evaluation use deterministic resizing and normalization.

Table 1. PlantVillage split used in the final experiments.

Split	Images Fraction	
Train	30,412	70%
Validation	6,516	15%
Test	6,516	15%

All final runs use pretrained ImageNet weights, batch size 32, 15 epochs, weight decay 10^{-4} , and Apple Silicon MPS as the selected device. The ResNet50 teacher uses learning rate 3×10^{-4} in the repository configuration, while the MobileNetV3-Small supervised baseline uses learning rate 10^{-3} . The distillation run uses learning rate 5×10^{-4} , temperature 4.0, and $\alpha = 0.5$. The best validation checkpoint is used for final test evaluation.

Table 2. Model and training configuration for the final experiments.

Model	Role	Learning rate	Epochs
ResNet50	Teacher supervised	3×10^{-4}	15
MobileNetV3-Small	Student supervised	10^{-3}	15
MobileNetV3-Small	Distilled student	5×10^{-4}	15

Table 3 reports the best validation checkpoints selected during training. The teacher’s best validation checkpoint was saved at epoch 9, with validation loss 0.0189, validation accuracy 0.9955, and validation macro-F1 0.9940. The supervised MobileNetV3-Small baseline reached its best validation accuracy at epoch 14, while the distilled student reached its best validation accuracy at the final epoch. This is consistent with the distillation run continuing to benefit from teacher supervision late in the 15-epoch schedule.

Table 3. Best validation checkpoints selected during training.

Model	Epoch	Val. loss	Val. acc.	Val. macro-F1
ResNet50 teacher	9	0.0189	0.9955	0.9940
MobileNetV3-Small supervised	14	0.0518	0.9850	0.9794
MobileNetV3-Small distilled	15	0.0216	0.9945	0.9931

Inference latency is reported from the repository’s batch-one timing routine, which averages repeated forward passes on the selected device. The latency values should therefore be interpreted as local Apple MPS measurements rather than hardware-independent deployment benchmarks.

5 Results and Discussion

Table 4 summarizes the final test-set results. The ResNet50 teacher reaches 0.9936 test accuracy and 0.9912 macro-F1. The supervised MobileNetV3-Small baseline is already strong, with 0.9851 accuracy and 0.9779 macro-F1. After distillation from ResNet50, MobileNetV3-Small improves to 0.9937 accuracy and 0.9907 macro-F1 while retaining the same 1.56M parameter count.

Table 4. Final test results on the 38-class PlantVillage task.

Model	Training strategy	Accuracy	Macro-F1	Parameters	Latency
ResNet50	Teacher supervised	0.9936	0.9912	23,585,894	5.47 ms
MobileNetV3-Small	Supervised baseline	0.9851	0.9779	1,556,806	3.51 ms
MobileNetV3-Small	Distilled from ResNet50	0.9937	0.9907	1,556,806	3.53 ms

The distilled student uses approximately 15.1 times fewer parameters than the teacher while matching it closely on accuracy and macro-F1. The latency of the distilled student remains close to the supervised MobileNetV3-Small baseline and below the measured ResNet50 latency in this local setup. These results suggest that, for this dataset and split, distillation is effective not merely as a compression technique but also as a way to recover much of the teacher’s predictive behavior in the smaller architecture.

The comparison between the supervised and distilled MobileNetV3-Small models is particularly important because the architecture and parameter count are the same in both rows. The improvement from 0.9851 to 0.9937 accuracy therefore does not come from increasing student capacity. Instead, it indicates that the teacher logits provide useful training information beyond the hard labels alone. The macro-F1 improvement from 0.9779 to 0.9907 suggests that the gain is not limited to majority classes, although per-class behavior should still be inspected before drawing conclusions about deployment robustness.

A full 38×38 confusion-matrix heatmap is difficult to read after it is scaled into the LNCS text width. Table 5 therefore reports the largest off-diagonal entries instead. The supervised MobileNetV3-Small model makes 97 test errors across 59 non-zero off-diagonal cells, while the distilled student makes 41 errors across 22 such cells. The full heatmap PNGs are kept as repository artifacts for detailed inspection.

The remaining errors are concentrated in visually related categories, such as maize leaf-spot/blight classes and several tomato disease classes, where symptoms

Table 5. Largest off-diagonal confusion-matrix entries on the test split.

Model	True class	Predicted class	Count
Supervised	Corn gray leaf spot	Corn northern leaf blight	8
Supervised	Tomato late blight	Tomato early blight	5
Supervised	Tomato spider mites	Tomato mosaic virus	4
Supervised	Tomato target spot	Tomato septoria leaf spot	4
Supervised	Orange HLB	Corn healthy	3
Distilled	Corn gray leaf spot	Corn northern leaf blight	9
Distilled	Tomato early blight	Tomato target spot	4
Distilled	Tomato yellow leaf curl virus	Tomato bacterial spot	4
Distilled	Apple black rot	Apple healthy	3
Distilled	Tomato target spot	Tomato spider mites	3

can be similar at image level. This pattern supports using macro-F1 alongside accuracy, since class-level behavior matters in a multi-class plant health setting.

The comparison should still be interpreted with caution. PlantVillage images are relatively clean and controlled, which can make the task easier than in-field plant disease recognition. The high supervised student result supports this point: MobileNetV3-Small already performs well before distillation. Therefore, the most defensible conclusion is that distillation preserves very high accuracy in a much smaller model on PlantVillage, not that the model is ready for unconstrained agricultural deployment.

Parameter counts clarify the deployment motivation. In the final 38-class configuration, ResNet50 has 23,585,894 trainable parameters, whereas the compact student has 1,556,806. The distilled student therefore keeps the smaller model size of the supervised student while approaching the teacher’s predictive performance. In this setting, it is not meaningfully slower than the supervised student, with measured latency of 3.53 ms compared with 3.51 ms. This small difference is expected because the teacher is used only during training; inference uses the student alone.

6 Reproducibility and Limitations

The final experiments are configuration-driven. The teacher run corresponds to the ResNet50 full-dataset configuration, the supervised student run corresponds to the MobileNetV3-Small full-dataset configuration, and the distillation run uses the ResNet50 teacher checkpoint saved under the full teacher output directory. The reported test metrics are taken from the corresponding result files in the repository outputs rather than copied from terminal logs. This keeps the report tied to reproducible artifacts: each run stores a model summary, training history, final metrics, a single-row result CSV, and a confusion matrix.

Several limitations remain. First, the split is deterministic but generated from the available PlantVillage directory rather than from an official external benchmark split. Second, PlantVillage leaf images are comparatively controlled;

performance on field images with cluttered backgrounds, lighting variation, occlusion, and different cameras may be lower. Third, the latency measurements are local Apple Silicon MPS measurements. They are useful for comparing the models in this environment, but they should not be interpreted as measurements from a deployed mobile application or embedded accelerator.

The distillation hyperparameters were also kept fixed in the final comparison. The selected temperature $T = 4.0$ and weight $\alpha = 0.5$ are reasonable defaults and worked well in this setting, but a more complete study would vary temperature, loss weighting, learning rate, and training length. Such tuning would help separate the effect of the teacher architecture from the effect of the distillation schedule.

7 Conclusion

This work evaluated knowledge distillation for edge-efficient plant disease classification on the full 38-class PlantVillage dataset. A ResNet50 teacher was distilled into a MobileNetV3-Small student, producing a compact model with 0.9937 test accuracy and 0.9907 macro-F1. The student has 1.56M parameters, about 15.1 times fewer than the 23.59M-parameter teacher, and maintains low local batch-one latency.

The results indicate that knowledge distillation can preserve near-teacher performance in a substantially smaller model in this controlled setting. The main limitation is dataset realism: PlantVillage is cleaner than many field images, so future work should evaluate on more realistic plant disease datasets, test on actual edge hardware, and tune the distillation temperature and loss weighting more systematically.

References

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